

Skills Summary

Terms, Factors, Coefficients, and Degree

Use this reference while completing your worksheet or when studying.

Skill 1 — Read an area model as terms

Model → **notation**:

Side lengths are **factors**; each cell inside is one **term** of the product. Cells with the same power of x are like terms and combine, so a four-cell model usually gives three terms.

$$(x + p)(x + q) = x^2 + (p + q)x + pq$$

Example: Expand $(x + 2)(x + 4)$ with a model.

Step 1. Each side is a factor, so I multiply every part of one side by every part of the other and read off the cells.

Step 2. Cells: x^2 , $4x$, $2x$, 8 . The middle two are like terms ($4x + 2x = 6x$): $x^2 + 6x + 8$

Try it: Expand $(x + 1)(x + 7)$.

$$x^2 + 8x + 7$$

Skill 2 — Name the factors of a polynomial

Rule: Factoring rewrites a *sum of terms* as a *product of factors* — the same value, a different form.

Always look for a common factor first: every term of $ax^m + bx^n$ shares the greatest common numerical factor and the lowest power of x . Degrees of factors **add** to the degree of the product.

Example: Factor $x^2 + 9x + 20$.

Step 1. The terms share no common factor, so I look for two numbers whose product is the constant term and whose sum is the middle coefficient.

Step 2. $4 \cdot 5 = 20$ and $4 + 5 = 9$, so $(x + 4)(x + 5)$ — two factors of degree 1, product of degree 2.

Try it: Factor $4x^2 - 6x$ completely (pull out the greatest common factor).

$$2x(2x - 3)$$

Skill 3 — Coefficient and degree

Vocabulary on one term, $-7x^3$: coefficient -7 (the sign belongs to it), variable x , degree 3 (the exponent).

Polynomial: its **degree** is the largest term degree; its **leading coefficient** is the coefficient of that highest-degree term — found by writing the polynomial in standard form (degrees decreasing), not by reading the first term as typed. A constant has degree 0; x alone has coefficient 1.

Example: Put $5 - 2x^2 + 7x^3$ in standard form; give its degree and leading coefficient.

Step 1. Leading coefficient is defined from standard form, so I reorder the terms by degree before reading anything off.

Step 2. Degrees are 0, 2, 3, so descending order gives $7x^3 - 2x^2 + 5$ — degree 3, leading coefficient 7.

Try it: Write $3x - 8 + x^4$ in standard form. What is its leading coefficient?

check: $x^4 - 8 + 3x$, leading coefficient 1

Skill 4 — Same expression, two forms (equivalent-forms)

Rule: Two expressions are **equivalent** when they give the same value for *every* x — proved by algebra (distributing or factoring), never by checking a few inputs in a table.

Counting terms does not decide it: $3(2x + 5)$ is one term made of two factors, and $6x + 15$ is two terms; they are the same expression written two ways.

Example: Show $2(3x - 4)$ as a sum of terms.

Step 1. The expression is a product, so distributing the outside factor turns it into a sum without changing its value.

Step 2. $2 \cdot 3x = 6x$ and $2 \cdot (-4) = -8$, giving $6x - 8$ — one term became two, the value did not change.

Try it: Expand $(x + 5)(x - 5)$. How many terms does the answer have?

check: $x^2 - 25$, two terms